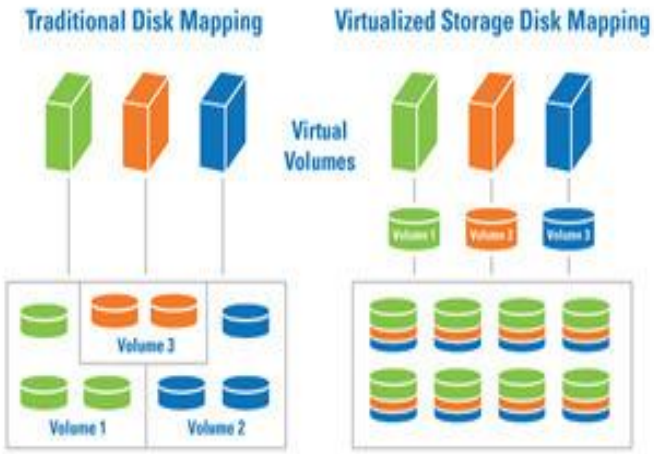


CLOUD COMPUTING COURSE

Lecture24- Virtualized Storage

ABSTRACT

Lecture24 Introduces virtualized storage in cloud.

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Virtualized Storage

Storage Virtualization

Storage virtualization is **the process of grouping the physical storage from multiple network storage devices so that it looks like a single storage device.**

The process involves **abstracting and covering** the internal functions of a storage device from the host application, host servers or a general network in order to facilitate the application and network-independent management of storage.

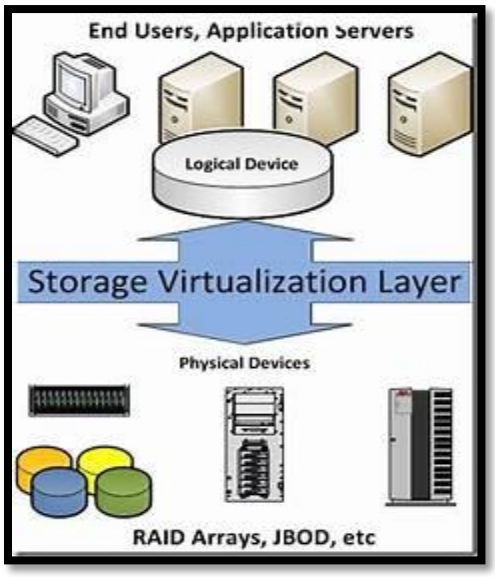
Storage virtualization can be implemented by using software applications or appliances.

The management of storage and data is becoming difficult and time consuming. Therefore, the **storage virtualization** helps to:

1. Improved storage management in a heterogeneous IT environment.
2. Better availability and estimation of down time with automated management
3. Better storage utilization

Storage virtualization can be applied to:

1. Any level of a SAN.
2. A physical storage.
3. RAID groups.
4. Logical unit numbers (LUNs) and logical volumes.



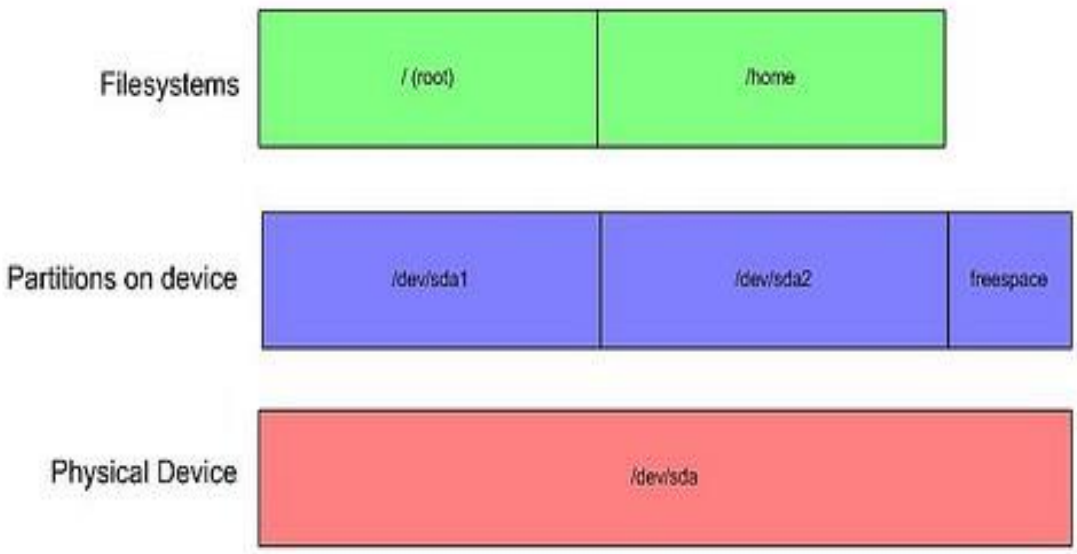
Visualized Storage Creation

We know that **physical disks** reside at the bottom of the storage system, either centralized or distributed. After RAID or replication is implemented, **physical volumes** are created on top of these physical disks.

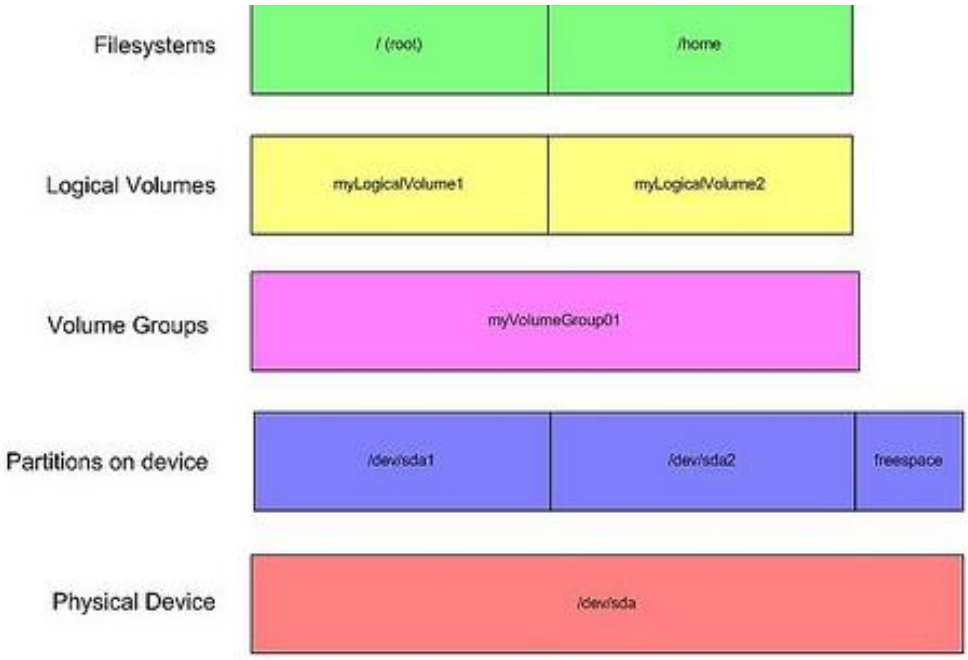
In most cases, the physical volumes are not directly mounted to upper-layer applications, for example, OSs or virtualization systems.

Therefore, typically, multiple physical volumes are combined into a **volume group**, and then the volume group is virtualized into multiple **logical volumes (LVs).** The upper-layer applications use the spaces of the **LVs.**

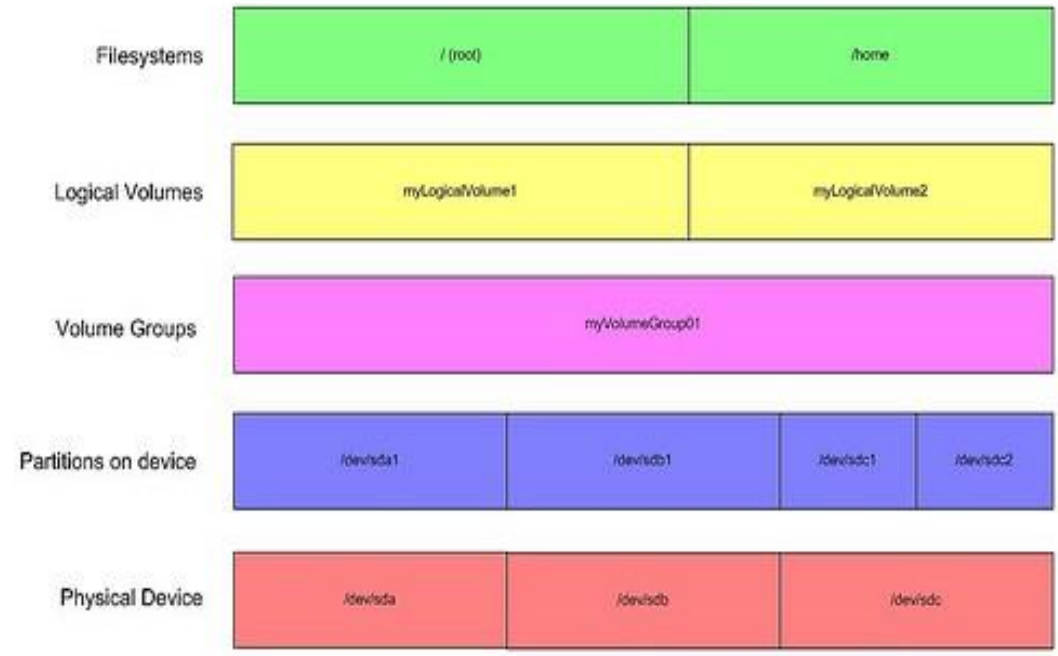
1. **Old Storage System**



2. Virtualized Storage



3. Adding More Disk to Virtualized Storage



Watch the related video: **https://youtu.be/JnAJQ3dS0fs**